Diwen Si

Robot Design, Mechatronics and Control Researcher

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Website | Ballbot project and videos | linkedin.com/in/diwen-si-944358220

Research Profile

Final-year PhD candidate in Mechanical Engineering at Newcastle University, completing a thesis on robot morphology, mechatronic sensing, system identification and learning-based control (thesis submission November 2026; viva expected January 2027). Designed, fabricated and experimentally commissioned a compact three-wheel Mecanum ballbot, validated control interventions on the physical robot, and built reproducible Webots and MuJoCo workflows for morphology–control co-design, command-following and controller evaluation. Five first-author manuscripts in preparation. Interested in co-optimising robot designs and control policies through simulation and validating them on physical hardware.

Research Interests

Morphology–control co-design; robot morphology and nonlinear mechanics; system identification; motor control; learning-based and model-based control; simulation-to-hardware validation.

Education

PhD Mechanical Engineering, Newcastle University, UK 2020 – Present

Thesis: *A Generalised Kinematic Framework for Ballbot Design: Optimisation, Prototyping and Learning-Based Control*. Submission November 2026; viva expected January 2027. Supervisors: Dr John Hedley and Dr Dehong Huo.

MSc Mechanical Engineering (Distinction), Newcastle University, UK 2018 – 2019

Dissertation: *Developing an Active Actuation Method for the Control of a Lower-Limb Exoskeleton*: a 10-DOF exoskeleton model with DH-based kinematics, centre-of-mass and ZMP analysis, inverse kinematics and gait trajectory generation.

BSc Mechanical Engineering, Chongqing Jiaotong University, China 2014 – 2018

Research Experience

Doctoral Researcher – Robot Morphology and Mecanum Ballbot Control 2022 – Present

School of Engineering, Newcastle University, UK

- Developed a generalised kinematic (actuation-Jacobian) framework for arbitrary driven-wheel count, contact placement and roller angle, combining rank, singular-value, conditioning, task-direction and redundancy analyses.
- Derived the exact isotropic geometry for equally spaced three-wheel Mecanum layouts (contact elevation 35.26°), built the prototype at a nominal 35°, and quantified installation-error sensitivity: roller-angle errors perturb the transmission map about twice as much per degree as contact-elevation errors.
- Designed, fabricated and integrated a low-centre-of-mass three-wheel Mecanum ballbot with motors, drivers, IMU, encoders, Raspberry Pi and an Arduino; synchronised command, inertial, current and external-pose telemetry for repeatable validation.
- Demonstrated closed-loop driving on hardware: 20 hand-driven runs (36 minutes in all, longest 480 s) and target following with an onboard Pixy2 camera.
- Ran a matched 36-run hardware comparison (18 pairs) on MK5: a separate tilt-correction map returned the robot upright in 18 of 18 trials, against 3 of 18 for a shared map; median tilt RMS fell from 14.2° to 4.9° and measured current norm from 6.53 to 3.54 A.
- Built a digital twin of MK5 from onboard logs: replaying recorded robot commands in simulation reduced the wheel-speed error from 20.6 to 7.8 r/min and raised the correlation from 0.581 to 0.895; fitting the actuator timing reduced the error from 15.9 to 7.5 r/min.
- Built Webots and MuJoCo models and reproducible experiment pipelines, including a 2,160-run wheel-construction benchmark and a 2,700-episode goal-conditioned reinforcement-learning study.
- Trained and evaluated SAC controllers, with and without hindsight experience replay (HER), under fixed action, observation, reward and evaluation protocols.
- Built a 672-body MuJoCo family of ballbots: one analytic controller closed 69% of the gap between doing nothing and perfect tracking on a separate set of 224 evaluation bodies when each body supplied its transmission map, against 18% with a nominal map.
- Developed recurrent controllers that read action–response history; on 36 bodies the final recurrent controller tracked better than zero action and than the analytic controller without body parameters. Developed a matched zero-action protocol for finite-horizon control benchmarks.
- Five builds led to the MK5 prototype; kept analysis code, configurations and figures versioned so that results can be regenerated.

Early PhD Research – Bipedal Robot Modelling and Control 2020 – 2022

School of Engineering, Newcastle University, UK

- PhD proposal on bipedal robots: dynamics, stability and simulation-driven controller studies. After the first supervisor left the University, moved to Dr Hedley's group and refocused the PhD on the ballbot platform.

Manuscripts in Preparation

- **Si, D.**, Hedley, J., and Wu, H. *Canonical Transmission Maps for Task-Specialised Morphology Design of Multiwheel Ballbots*. In preparation for *Mechanism and Machine Theory*.
- **Si, D.** and Hedley, J. *Shared Actuators Need Not Share Coordinates: Role-Factorized Allocation on a Mecanum Ballbot*. In preparation for *IEEE/ASME Transactions on Mechatronics*.
- **Si, D.** and Hedley, J. *When Full-Rank Geometry Is Not Enough: Body-Local Response Commissioning Across Ballbot Morphologies*. Manuscript in preparation.
- **Si, D.** and Hedley, J. *Goal Representation and Hindsight Replay for Webots Ballbot Control: A Controlled Factorial Study*. In preparation for *Robotics and Autonomous Systems*.
- **Si, D.** *Doing Nothing Scores Well: A Paired Zero-Action Protocol for Auditing Finite-Horizon Robot Control Benchmarks*. Manuscript in preparation.

Teaching and Academic Experience

Graduate Teaching Assistant, Newcastle University 2021 – Present

- Supported MEC3027 Instrumentation and Drives, MEC3033 Introduction to Mechatronics Design, MEC8057 Mechatronics and Mobile Robots, and Applied AI and Robotics at postgraduate and undergraduate levels.
- Guided students in Python/MATLAB control and sensing exercises, embedded platforms and laboratory debugging; marked video-based assignments against the module template and explained robotics concepts through practical examples.
- Guided undergraduate teams building line-following robot cars, and helped Dr Harriet Grigg supervise an MSc dissertation on an exoskeleton.

Professional recognition: Associate Fellow of Advance HE (AFHEA), Newcastle University.

Technical Skills

Programming and data	Python, NumPy, pandas, Matplotlib, Jupyter, MATLAB, experiment scripting, signal and time-series analysis, sensor-log synchronisation, CSV/JSON workflows
Robotics and control	Webots, MuJoCo, Deepbots/Gym, Stable-Baselines3/PyTorch, SAC, HER, recurrent policies, kinematics, motor control, perception/telemetry evaluation, system identification
Hardware and embedded	Raspberry Pi, Arduino, Maxon ESCON servo drivers, motor control, IMU/encoder integration, motor-current and AprilTag/video pose logging, electronics, wiring, safety and hardware commissioning
Engineering tools	SolidWorks, Fusion 360, AutoCAD, Git, Linux, LaTeX, VS Code
Languages	Mandarin Chinese (native); English (fluent)